- **Random variable**
  A random variable X is a measurable function that maps the sample space Ω into the real numbers $\mathbb{R}$:

$$X : \Omega \to \mathbb{R}, \quad \text{such that } \forall a \in \mathbb{R}, \{\omega \in \Omega : X(\omega) \leq a\} \in \mathcal{F}$$

- **Discrete random variable**
  A random variable X is called discrete if its range $S_X \subset \mathbb{R}$ is finite or countably infinite, i.e., $S_X = \{x_1, x_2, \dots\}$.

- **Probability mass function (pmf)**
  The probability mass function $p_X(x)$ for a discrete random variable X is a function $p_X : \mathbb{R} \to \mathbb{R}$ such that :

$$p_X(a) = P(X = a)$$

- **Cumulative distribution function (cdf)** for a random variable X is a function $F_X : \mathbb{R} \to \mathbb{R}$ such that :

$$F_X(a) = P(X \leq a)$$

- **Bernoulli distribution**
  $X \sim \mathscr{B}(1, p)$ with $p \in (0, 1)$ iff it's pmf is :

$$p_X(k) = \begin{cases} p, & k = 1 \\ 1 - p, & k = 0 \end{cases}$$

- **Binomial distribution**
  $X \sim \mathscr{B}(n, p)$ with $p \in (0, 1), n \in \mathbb{N}$ iff it's pmf is :

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k \in \{0, 1, 2, \dots, n\}$$

- **Geometric distribution**
  $X \sim \mathscr{G}(p)$ with parameter $p \in (0, 1)$ iff it's pmf is :

$$p_X(k) = (1-p)^{k-1} p, \quad k \in \{1, 2, 3, \dots\}$$

- **Poisson distribution**
  $X \sim \mathscr{P}(\lambda)$ with parameter $\lambda > 0$ iff it's pmf is :

$$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k \in \{0, 1, 2, \dots\}$$

- **Joint probability mass function**
  The joint pmf for two discrete random variables X and Y gives the probability of their simultaneous occurrence:

$$p_{X,Y}(x, y) = P(X = x \text{ and } Y = y) = P(X = x, Y = y)$$

- **Marginal probability mass function**
  The marginal pmf of a random variable X is obtained by summing the joint pmf over all possible values of Y:

$$p_X(x) = P(X = x) = \sum_y p_{X,Y}(x, y)$$

- **Conditional distribution**
  The conditional probability mass function of X given that $Y = y$ (where $p_Y(y) > 0$) is defined as:

$$p_{X|Y}(x \mid y) = P(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

- **Independent random variables**
  Random variables X and Y are independent if

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y))$$

- **Expected value**
  The expected value (mean) of a discrete random variable X is defined as:

$$E[X] = \mu_X = \sum_{x \in S_X} x \cdot p_X(x)$$

  (provided that $\sum_x |x| p_X(x) < \infty$).

- **Properties of expectation**

1. Linearity: $E[aX + bY + c] = aE[X] + bE[Y] + c$, for constants $a, b, c \in \mathbb{R}$
2. $E[g(X)] = \sum_{x \in S_X} g(x) \cdot p_X(x)$

- **Variance**
  The variance $\mathrm{Var}(X)$ of a random variable X:

$$\mathrm{Var}(X) = E\left[(X - E[X])^2\right] = E[X^2] - (E[X])^2$$

- **Properties of variance**

1. Non-negativity: $\mathrm{Var}(X) \geq 0$; $\mathrm{Var}(c) = 0$ for any constant c
2. Shift and scale: $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$
3. Sum of variables: $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$
4. For independent variables X, Y: $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$

- **Covariance**
  Covariance of X and Y:

$$\mathrm{Cov}(X, Y) = E\left[(X - E[X])(Y - E[Y])\right] = E[XY] - E[X]E[Y]$$

- **Correlation coefficient**
  Correlation coefficient between X and Y is:

$$\mathrm{Corr}(X, Y) = \rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

- **Probability generating function (p.g.f.)**
  The probability generating function $G_X(s)$ for an integer-valued random variable X is the function of real variable defined by:

$$G_X(s) = E[s^X] = \sum_k p_X(k)s^k, \quad s \in \mathbb{R}$$

- **Properties of p.g.f.**

1. $p_X(k) = \frac{G_X^{(k)}(0)}{k!}$
2. $\mu^{(k)} = E[X(X-1)\dots(X-k+1)] = G_X^{(k)}(1)$
3. Sum of independent variables: If X and Y are independent, then $G_{X+Y}(s) = G_X(s) \cdot G_Y(s)$